

Practical Research 2

Quarter 2 - Module 4

Understanding Data and Ways to Systematically Collect Data

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Practical Research 2- Grade 12

Alternative Delivery Mode

Quarter 2 – Module 4

Understanding Data and Ways to Systematically Collect Data

First Edition, 2020

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Senior High School

Practical Research 2

Quarter 2- Module 4 Understanding Data and Ways to Systematically Collect Data

This instructional material was collaboratively developed and reviewed by educators from public and private schools, colleges, and or/universities. We encourage teachers and other education stakeholders to email their feedback, comments, and recommendations to the Department of Education at action@deped.gov.ph.

We value your feedback and recommendations.

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What This Module is About

When you aim to arrive to a certain solution of a problem, the journey is definitely smooth and focused when the direction is clear. The research methodology of your study is very important since it gives direction to systematically arrive to answers or solution to your chosen research problem. It contains rules and procedure that you as a researcher will have to follow.

This module focuses on the discussion of your research design and plan, population to consider, sampling technique, research instruments, and appropriate statistical treatments to be employed. At end of this learning module learners are expected to understand the concept of research data and know the ways to systematically collect data.

The following are the lessons contained in this module:

- Lesson 1 – Quantitative Research Design
- Lesson 2 – Sampling Procedure and Sample
- Lesson 3 – Research Instruments, Validity, Reliability
- Lesson 4 – Research Intervention
- Lesson 5 – Data Collection Procedure
- Lesson 6 – Data Analysis
- Lesson 7 – Presenting Research Methodology



What I Need to Know

At the end of this module, you should be able to:

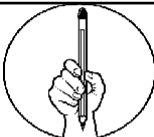
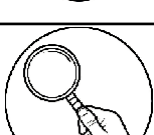
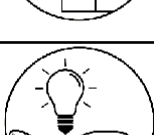
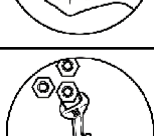
1. chooses appropriate quantitative research design (**CS_RS12-Ila-c-1**);
2. describes sampling procedure and the sample (**CS_RS12-Ila-c-1**);
3. constructs an instrument and establishes its validity and reliability (**CS_RS12-Ila-c-3**);
4. describes intervention (if applicable) (**CS_RS12-Ila-c-4**);
5. plans data collection procedure (**CS_RS12-Ila-c-5**);
6. plans data analysis using statistics and hypothesis testing (if appropriate) (**CS_RS12-Ila-c-6**);
7. presents written research methodology (**CS_RS12-Ila-c-7**);

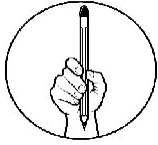
How to Learn from this Module

To achieve the objectives cited above, you are to do the following:

- Take your time reading the lessons carefully.
- Follow the directions and/or instructions in the activities and exercises diligently.
- Answer all the given tests and exercises.

Icons of this Module

	What I Need to Know	This part contains learning objectives that are set for you to learn as you go along the module.
	What I know	This is an assessment as to your level of knowledge to the subject matter at hand, meant specifically to gauge prior related Knowledge
	What's In	This part connects previous lesson with that of the current one.
	What's New	An introduction of the new lesson through various activities, before it will be presented to you
	What is It	These are discussions of the activities as a way to deepen your discovery and understanding of the concept.
	What's More	These are follow-up activities that are intended for you to practice further in order to master the competencies.
	What I Have Learned	Activities designed to process what you have learned from the lesson
	What I can do	These are tasks that are designed to showcase your skills and knowledge gained, and applied into real-life concerns and situations.



What I Know

Directions: Read and analyze the statements below. Encircle the letter of the correct answer.

1. It is a research design that aims to determine a cause from already existing effects.
A. Descriptive Research Design C. Ex Post Facto
B. Correlational Research Design D. Quasi-Experimental Research
2. It is a research design that is often conducted in a controlled setting with corresponding research treatment.
A. Correlational C. Survey Research
B. Ex post facto D. Experimental
3. What is the suited research design for this research title, "The Effects of Twitter on the Communication Etiquette of Students"?
A. Correlational C. Descriptive
B. Ex post facto D. Experimental
4. Mr. Canuc would like to further know the type of social media used between the male and female SHS students of East Pagat National High School. What is the appropriate research design to be used in his study?
A. Descriptive C. Quasi-Experimental
B. Correlational D. Experimental
5. What is the difference between quasi-experimental research and experimental research?
A. Intact groups are used in experimental, while quasi-experimental randomly assigned individuals into groups.
B. The researcher controls the intervention in the experimental group in experimental, but not quasi-experimental research.
C. Only one dependent variable is used in quasi-experimental research, while multiple dependent variables can be used in quasi-experimental research.
D. Participants for control and experimental groups are randomly selected in experimental, but not quasi-experimental research.
6. It is used when the researcher would like to consider giving equal chance to the member of accessible population being selected as part of the study.
A. Simple Random Sampling C. Cluster Sampling
B. Stratified Sampling D. Systematic Sampling
7. The research sample is considered best when it is
A. purposely selected C. representative of population
B. systematically chosen D. conveniently represented
8. Stratified random sampling ensures that
A. those who will possibly respond to treatment are chosen
B. sample chosen proportionately drawn from the different categories of population
C. every individual will be given an equal chance to be selected
D. sample is taken from accessible population than target population

9. Identify the sampling method used in the given situation.

Teacher Joan wants to know if the new learning modalities of the school have an effect on the academic performance of students in science curriculum. He took the list of students and selected every 8th name in each class list as participant.

- | | |
|-------------------------------|-------------------------------|
| A. Stratified Random Sampling | C. Systematic Random Sampling |
| B. Simple Random Sampling | D. Cluster Sampling |

10. It is a type of validity when an instrument produces results similar to those of another instrument that will be employed in the future.

- | | |
|------------------------|---------------------|
| A. Predictive Validity | C. Face Validity |
| B. Criterion Validity | D. Content Validity |

11. It is a type of reliability when measured by administering two tests identical in all aspects except the actual wording of items.

- | | |
|-------------------------------------|---------------------------------|
| A. Test-retest reliability | C. Equivalent Forms Reliability |
| B. Internal Consistency Reliability | D. Inter-rater Reliability |

12. The Ability Test has been proven to predict the writing skills of Senior High School students is an example of

- | | |
|-----------------------|---------------------|
| A. Construct Validity | C. Content Validity |
| B. Criterion Validity | D. Face Validity |

13. It is a common scaling technique which consists of several declarative statements that express viewpoint on a topic.

- | | |
|--------------------------------|--------------------------|
| A. Semantic Differential Scale | C. Likert Scale |
| B. Completion Type | D. Observation Checklist |

14. The purpose of this statistical technique is to test the relationship between two variables.

- | | |
|--------------------|-------------------------------------|
| A. Chi Square Test | C. T-Test for two dependent samples |
| B. Pearson's r | D. T-Test for independent samples |

15. What statistical technique should be used for this research question, "Is there a significant difference between the pretest and posttest scores of learners in reading comprehension test?"

- | | |
|--------------------|-------------------------------------|
| A. Chi Square Test | C. T-Test for two dependent samples |
| B. Pearson's r | D. T-Test for independent samples |

Lesson 1

Quantitative Research Designs



What's In

Quantitative research is basically more systematic and controlled than qualitative research. However, both research methods have statement of the problem to investigate. At this point, it is assumed that you are already done stating your research problem, background of the study, scope and delimitation, hypothesis, conceptual framework, and review of related literature and studies.

Each type of research has unique design to follow that will also lead the direction of sampling procedure, data gathering, and data analysis. Each research type also aims to answer specific research questions, how it will be answered is determined by its design.

Important considerations also are the variables of the study. The type of variables provides paradigm (your conceptual framework) which helps researcher decides on what will be the design of the study.

So it is very important, that you have completed the previous tasks diligently in order to maximize your learning in this Module. In this lesson, you will be taught how to select appropriate quantitative research design.



What I Need to Know

Meaning of Quantitative Research Designs

When a man decides to build a house, does not he draws first the blue print before he will start to do the work? Same with the conduct of research, the blueprint for collection, measurement, and data analysis is drawn as a pattern to follow. Furthermore, research designs enable the researcher to obtain a more valid, objective, reliable, and accurate answers to the research questions.

Research design is define as the logical and coherent overall strategy that the researcher uses in order to integrate all the components of the research study (Barrot, 2017, p 102). In order to find meaning in the overall process of doing your research study, a step-by-step process will be helpful to you.

In quantitative research, you are going to have a great deal of abstraction and numerical analysis. According to Fraenkel and Wallen (2007, p 15), the research designs in quantitative research are mostly pre-established. Hence having an appropriate research design in quantitative research, the researcher will have a clearer comprehension of what you are trying to analyze and interpret.



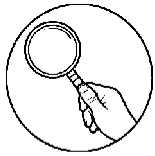
What's New

Activity 1: Plan Your Work

Directions: Pretend that you are conducting a study on academic adjustments of teachers and students on the learning delivery mode during this time of COVID – 19 pandemic. Come up with a plan on how you are going to obtain necessary data for this study. Draw a procedural flowchart to illustrate your plan.

Study on Academic Adjustments of Teachers and Students
on the Learning Delivery Modes during COVID – 19 Pandemic

“Plan of Work Flowchart”



What Is It

Types of Quantitative Research Design

Quantitative Research Designs have five general classifications: descriptive, correlational, ex post facto, quasi-experimental, and experimental.

Descriptive Research. When little is known about the research problem then it is appropriate to use descriptive research design. It is a design that is exploratory in nature. The purpose of descriptive research is basically to answer questions such as “who”, “what”, “where”, “when”, and “how much”. So this design is best used when the main objective of the study is to just observe and report a certain phenomena as it is happening.

Correlational Research. The main goal of this design is to determine if a variable increases or decreases as another variable increases or decreases. This design seeks to establish association between variables. It does not seek cause and effect relationship like descriptive research it measures variables as it occur. It has two major purposes: (a) to clarify relationship between variables and (b) predict the magnitude of association. However, the extent of purpose of correlational research depends on the scope and delimitation of the study.

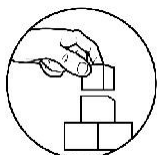
Ex Post Facto. If the objective of the study is to measure a cause from a pre existing effects, then Ex Post Facto research design is more appropriate to use. In this design, the researcher has no control over the variables in the research study. Thus, one cannot conclude that the changes measured happen during the actual conduct of the study.

The last two types of quantitative research designs are identifiable for the existence of treatment or intervention applied to the current research study. **Intervention or treatment** pertains to controlling or manipulating the independent variable in an experiment. It is assumed that the changes in dependent variables were caused by the independent variable.

There are also two groups of subjects, participants or respondents in quasi-experimental and experimental research. **Treatment group** refers to the group subjected to treatment or intervention. The group not subject to treatment or intervention is called **control group**.

Quasi-Experimental. The term means partly, partially, or almost – pronounced as *kwahz-eye*. This research design aims to measure causal relationship between variables. The effect measured is considered to have occurred during the conduct of the current study. The partiality of quasi-experimental design comes from assigning of subjects, participants, or respondents into their groups. The groups are known to be already established before the study such age educational background and nationality. Since assignment of subjects, participants, or respondents are not randomly assigned into experimental or control group, the conclusion of results is limited.

Experimental Research. This research design is based on scientific method called experiment with a procedure of gathering data under a controlled or manipulated environment. It is also known as true experimental design since it applies treatment and manipulation more extensively compared to quasi-experimental design. Random assignment of subjects or participants into treatment and control group is done; increasing the validity of the study. Experimental research, therefore, attempts to affect a certain variable by directly manipulating the independent variable.



What's More

Activity 2. Quantitative Research Designs Summary

Directions: Using the template below, summarize the five quantitative research designs according to its goal, and its corresponding variable manipulation.

Research Design	Goal	Variable Manipulation
1. Descriptive		
2. Correlational		
3. Ex post facto		
4. Quasi-experimental		
5. Experimental		

Activity 2. Choose the Appropriate Research Design

Directions: Determine the quantitative research design appropriate for the sample research titles. Make sure to explain your choice into two to three sentences.

1. Relationship between Academic Stressors and Learning Preferences of Senior High School Students

Quantitative Research Design: _____

Explanation:

2. Reading Electronic Learning Materials as a Support for Vocabulary of Grade 1 Pupils

Quantitative Research Design: _____

Explanation:

3. Impact of the Implementation of COVID – 19 Health Protocols in Supermarkets on Consumer Behaviors

Quantitative Research Design: _____

Explanation:

4. Effects of Morning Exercise on the Health Anxiety Level of Senior Citizens

Quantitative Research Design: _____

Explanation:

5. Measuring the Gadgets Usage of Children at Home during Quarantine

Quantitative Research Design: _____

Explanation:

6. Level of Academic Achievement of Senior High Schools in Different Learning Modalities

Quantitative Research Design: _____

Explanation:

7. Effects of Story Telling on Quality of Sleep of Children



Quantitative Research Design: _____

Explanation:

Activity 3: Let's go Online

Go to the link below and practice what you've learned from this lesson:

<https://bit.ly/36A5idj>
<https://bit.ly/2LZVLCS>
<https://bit.ly/3eojlWk>
<https://bit.ly/2X2kxbX>



What I Have Learned

Directions: As you have learned from this lesson, answer each question comprehensively.

1. This time, I have learned that quantitative research design....

2. How do you know the specific quantitative research design applied in a research study?

What I Can Do

Choose your Appropriate Research Design. Based on your own research title

Directions: Perform the following task. From what you have learned in this lesson, what is the appropriate research design for your current research problem? Justify your choice.

Research Design:

Justification:



Additional Activity:

Direction: After you submit your research design, once it is corrected in accordance to the guidelines given, you may start incorporating it to your research manuscript.

2

Sampling Procedure and the Sample



What's In

Remember when you were taught how to write your scope and delimitation, you stated the subjects, participants, or respondents of your study. You also described their characteristics which qualify them to be the source of your research data.

The next question you have to answer is, how many of the subjects, participants, or respondents should be selected as source of data? This lesson will teach you how to describe sampling procedure in quantitative research. Note that, the sampling procedure should be aligned to your chosen research design. Since you have already decided the research design of your study, then you are ready for this lesson.



What I Need to Know

Population and Sample

The first step in determining the sample size is identifying the population of the topic of interest. The **population** is the totality of all the objects, elements, persons, and characteristics under consideration. It is understood that this population possess common characteristics about which the research aims to explore.

There are two types of population: target population and accessible population. The actual population is the **target population**, for example, all Senior High School Students enrolled in Science, Technology, Engineering, and Mathematics (STEM) in the Division of Cagayan de Oro City. While the accessible population is the portion of the population in which the researcher has a reasonable access, for example all Senior High School enrolled, STEM strand at Gusa Regional Science High School – X.

When the whole population is too costly or time consuming or impractical to consider then a sample representative is identified. **Sampling** pertains to the systematic process of selecting the group to be analyzed in the research study. The goal is to get information from a group that represents the target population. Once good sample is obtained; the generalizability and applicability of findings increases.

The representative subset of the population refers to **sample**. All the 240 Senior High School Students enrolled in Science, Technology, Engineering, and Mathematics (STEM) Strand in a school for example constitute the population; 60 of those students constitute the sample. A good sample should have characteristics of the represented population – characteristics that are within the scope of the study with fair accuracy. Generally, the larger the sample the more reliable the sample be; but still it will depend on the scope and delimitation and research design of the study.



What's New

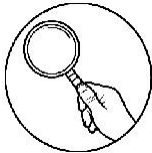
You are almost finished cooking the favourite meal of the family. In order to check if it tastes good and ready for serving, what will you do?

Answer:

I will.....



<https://bit.ly/2BfQXHJ>



What Is It

Approaches in Identifying the Sample Size

Heuristics. This approach refers to the rule of the thumb for sample size. Early established approach by Gay (1976) stated by Cristobal and Dela Cruz-Cristobal (2017, p 172), sample sizes for different research designs are the following:

Research Design	Number of Subjects/Participants
Descriptive Research	10% to 20% maybe required
Comparative Research	15 subjects or groups

Lunenberg and Irby (2008) as cited by Barrot (2017, p 107) also suggested different sample sizes for each quantitative research design.

Research Design	Number of Subjects/Participants
Survey	800
Correlational	100 to 200
Ex post facto	30+
Experimental	30 or more

Literature Review. Another approach is by reading similar or related literature and studies to your current research study. Since you are done writing your review of related literature and studies, you might want to recall how these studies determine sample size. Using this approach increases the validity of your sampling procedure.

Formulas. Formulas are also being established for the computation of acceptable sample size. Common formula is the Slovin's Formula.

9

Slovin's Formula:

$$n = \frac{N}{1 + Ne^2}$$

where: n is sample size
N is population size
E is desired margin of error

$$n = \frac{600}{1 + 600(0.05)^2}$$

$$i \frac{600}{1 + 600(0.0025)}$$

$$i \frac{600}{1 + 1.5}$$

analysis
approach

The probability of rejecting the null hypothesis is called **statistical power**. This suggests that indeed there is a relationship between the independent and dependent variables of the research study. The ideal statistical power of a research study is 80%. With the statistical power, it will be used to identify the sufficient sample size for measuring the effect size of a certain treatment. The level of difference between the experimental group and control group refers to **effect size**.

If the statistical power tells that relationship between independent and dependent variables, the effect size suggests the extent of relationship of these two variables. Henceforth, the higher the effect size, it means greater the level difference between the experimental and control groups. For example, your research study reveals that there is a difference in the pretest and posttest scores of the students in the given anxiety test after implementing psychosocial intervention. With the effect size, you will have an idea how small or large is the difference.

the principle of power
are going to use this

Useful online
tools:

Visit the
following links to
practice the power
analysis approach.

<https://bit.ly/371a3IC>

<https://bit.ly/2AQpDQn>

Probability Sampling in Quantitative Research

Simple Random Sampling. It is a way of choosing individuals in which all members of the accessible population are given equal chance to be selected. There are various ways of obtaining samples through simple random sampling. These are fish bowl technique, roulette wheel, or use of table of random numbers. This technique is also readily available online. Visit this link <https://www.randomizer.org/> to practice.

Stratified Random Sampling. The same with simple random sampling, stratified random sampling also gives equal chance to all members of the population to be chosen.

However, the population is first divided into strata or groups before selecting the samples. The samples are chosen from these subgroups and not directly from the entire population. This procedure is best used when the variables of the study are also grouped into classes such as gender and grade level.

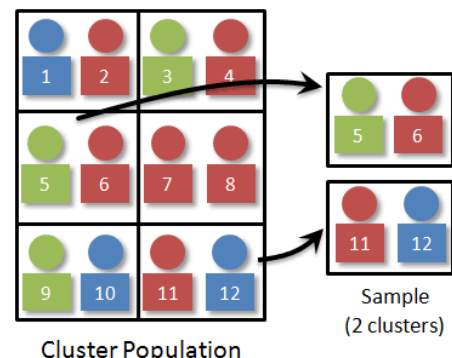
You can simply follow the steps from this given example: A population of 600 Junior High School students includes 180 Grade 7, 160 Grade 8, 150 Grade 9, and 110 Grade 10. If the computed sample size is 240, the following proportionate sampling will be as follows.

The number of members per subgroup is divided by the total accessible sample size. The percentage result of members per subgroup will be multiplied from the computed total sample size. After obtaining the sample size per strata, then simple random sampling will be done for the selection of samples from each group.

				Sample Size per Subgroups
180/600	=	.30x 240	=	72 Grade 7 students
160/600	=	.27x 240	=	65 Grade 8 students
150/600	=	.25x 240	=	60 Grade 9 students
110/600	=	.18x 240	=	43 Grade 10 students
100%				240 respondents

Cluster Sampling. This procedure is usually applied in large-scale studies, geographical spread out of population is a challenge, and gathering information will be very time consuming. Similar with stratified random sampling, cluster sampling also involves grouping of population according to subgroups or clusters. It is a method where multiple clusters of people from the chosen population will be created by the researcher in order to have homogenous characteristics.

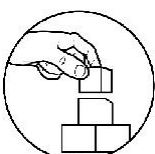
For example, a researcher would like to conduct an interview of all public senior high school students across Mindanao. As a researcher cluster will be selected to satisfy the plan size. In the given example, first cluster can be by region, second cluster can be by division, and third cluster can be by district.



Another way of doing cluster sampling is illustrated on the figure in the right side.

<https://bit.ly/2UTiZzp>

Systematic Sampling. This procedure is as simply as selecting sample every n^{th} (example every 2nd, 5th) of the chosen population until arriving at a desired total number of sample size. Therefore the selection is based on a predetermined interval. Dividing the population size by the sample size the interval will be obtained. For example, from a total population of 75 you have 25 samples, using systematic sampling you will decide to select every 3rd person on the list of individuals.



What's More

Activity 1. Determine the Sampling Procedure

Directions: Identify the sampling procedure used in each given situation. Write your answer on the space provided and then explain your choice.

Sample Situation	Sampling Procedure	Justification
1. Alex's target population for his study are the employees of hotels in Mindanao. Since there are too many employees in these establishments, he randomly selected ten hotels. And then he considered all employees as participants in his study.		
2. Dianne wants to know if the new learning modalities in first semester have an effect on the academic performance of senior high students. He took all the lists of all students in her school and selected every 6 th name to be part of her study.		
3. Faye wants to survey all the parents in Cagayan de Oro who opt to enrol their elementary children to online class. All in all there 26, 000 parents. Faye decided to have 450 from the target population.		

Activity 2. Search the design and sampling procedure.

Directions: Search for a sample research study. Identify the research design used and its sampling procedure.

	Example 1	Example 2	Example 3
▪ Title of the Research Study			
▪ Research Design			
▪ Characteristics of Population			
▪ Sampling Procedure			
▪ Sample Size			
▪ Source			

Activity 3. Let's go online.



Practice what you have learned from this lesson. Visit the following links:

<https://bit.ly/3em5Rdv>

<https://bit.ly/2TF09LO>

What I Have Learned



Directions: As you have learned from this lesson, answer each question comprehensively.

1. What I know about population and sample of the study?

2. I have learned that when determining the sample size of the study....

3. Now I know that sampling procedures involve.....

What I Can Do

Describes your Sampling Procedure and Sample

Directions: Perform the following task. Identify the size of population and sample for your study and explain the sampling method that you will use.

1. Size of Population	
2. Method used to determine sample size (include computation if applicable)	
3. Sample Size for the Study	
4. Sampling Procedure to be employed (explain the steps)	
5. Who will be your respondents?	



Additional Activity

Direction: After you submit you have describe your sampling procedure and sample, once it is corrected in accordance to the guidelines given, you may start writing a paragraph format of this and incorporate it in your research manuscript.

3

Research Instrument, Validity, and Reliability



What I Need to Know

Quantitative Research Instrument

What do you think will happen if tools for building a house is not prepare meticulously? Same thing when getting information for answers to research problem, tools or instrument should be prepared carefully. In constructing quantitative research instrument, it is very important to remember that the tools created should required responses or data that will be numerically analyzed.

Research Instruments are basic tools researchers used to gather data for specific research problems. Common instruments are **performance tests**, **questionnaires**, **interviews**, and **observation checklist**. The first two instruments are usually used in quantitative research while the last two instruments are often in qualitative research. However, interviews and observation checklists can still be used in quantitative research once the information gathered is translated into numerical data.



What's New

Activity 1. Let's talk

Directions: Read the questions about tools. Then, write your answer on the corresponding space.

Questions to analyze:

1. Why it is necessary to use right tools for a particular task?
2. What will happen if research instruments are not prepared carefully?

Answers:

1. _____

2. _____

Activity 2. Let's read

Directions: Read and analyze the given scenario. Answer the guide questions below.

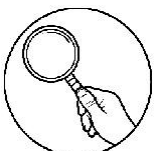
A culmination program was scheduled for Grade 12 students. The highlight of the program is the presentation of the festival of dances. The six sections prepared for the said dance contest during the culmination. A month before the activity the students already started planning. Their parents were also very supportive in the preparation of their costumes and props. The class advisers also monitored the practices in their classroom.

Guide Questions:

1. What do you think must have been done to avoid the said situation?

2. What can you say about the result of investigation?

3. How will you relate the scenario to the conduct of quantitative research study?



In constructing research instrument of the study there are many factors to be considered. The type of instrument, reasons for choosing the type, and the description and conceptual definition of its parts are some of the factors need to be decided before constructing research instrument. Furthermore, it is also very important to understand the concepts of scales of research instruments and how to established validity and reliability of instruments.

Characteristics of a Good Research Instrument

Concise. Have you tried answering a very long test and because of its length you just pick the answer without even reading it? A good research instrument is concise in length yet can elicit the needed data.

Sequential. Questions or items must be arranged well. It is recommended to arrange it from simplest to most complex. In this way, the instrument will be more favourable to the respondents to answer.

Valid and reliable. The instrument should pass the tests of validity and reliability in order to get more appropriate and accurate information.

Easily tabulated. Since you will be constructing an instrument for quantitative research this factor should be considered. Hence, before crafting the instruments, researcher makes sure that the variable and research questions are established. These will be an important basis for making items in the research instruments.

Ways in Developing Research Instrument

There are three ways you can consider in developing the research instrument for your study. First is **adopting an instrument** from already utilized instrument from previous related studies. Second way is **modifying an existing instrument** when the available instruments do not yield the exact data that will answer the research problem. And the third way is when the **researcher made his own instrument** that corresponds to the variable and scope of his current study.

Another important consideration in constructing research instrument is how to establish its validity and reliability.

Common Scales Used in Quantitative Research

Likert Scale. This is the most common scale used in quantitative research. Respondents were asked to rate or rank statements according to the scale provided.

Example: A likert scale that measures attitude of students towards distance learning.

Items	Strongly Agree	Agree	Disagree	Strongly Disagree
There would be difficulty in communicating our concerns to our teacher.				
There would be many distractions when learning at home than in school.				

Semantic Differential. In this scale, a series of bipolar adjectives will be rated by the respondents. This scale is seems to be more advantageous since it is more flexible and easy to construct.

Example: On a description of an active student in school activities.

Pleasant	5	4	3	2	1	Unpleasant
Enthusiastic	5	4	3	2	1	Not Enthusiastic
Competent	5	4	3	2	1	Incompetent

Types of Validity of Instrument

Face Validity. It is also known as “logical validity.” It calls for an initiative judgement of the instruments as it “appear.” Just by looking at the instrument, the researcher decides if it valid.

Content Validity. An instrument that is judged with content validity meets the objectives of the study. It is done checking the statements or questions if this elicits needed information. Experts in the field of interest can also provide specific elements that should be measured by the instrument.

Construct Validity. It refers to validity of instruments as it corresponds to theoretical construct of the study. It is concerned if specific measure relates to other measures.

Concurrent Validity. When the instrument is able to predict results similar to those similar tests already validated, it has concurrent validity.

Predictive Validity. When the instrument is able to produce results similar to those similar tests that will be employed in the future, it has predictive validity. This is particularly useful for aptitude test.

Validity

A research instrument is considered valid if it measures what it supposed to measure.

When measuring oral communication proficiency level of students, speech performance using rubric or rating scale is more valid than students are given multiple choice tests.

Validity has also several types: face, content, construct, concurrent, and predictive validity.

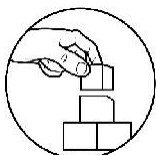
Reliability of Instrument

Reliability refers to the consistency of the measures or results of the instrument.

Test-retest Reliability. It is achieved by giving same test to same group of respondents twice. The consistency of the two scores will be checked.

Equivalent Forms Reliability. It is established by administering two identical tests except wordings to the same group of respondents.

Internal Consistency Reliability. It determines how well the items measure the same construct. It is reasonable that when a respondent get a high score in one item, he will also get one in similar items. There are three ways to measure the internal consistency; through split-half coefficient, Cronbach’s alpha, and Kuder-Richardson formula.



What’s More

Activity 1. Qualify and Rate the Instrument

Directions: Look for sample research instrument from previously conducted research. Rate it on the scale of 1 to 5 (1 = lowest, 5 = highest) based on the criteria given. Then justify your rating.

Sample Instrument (provide title of the instrument if possible):		
Purpose of the Research Instrument:		
Number of Items: _____	Scale Used: _____	
Criteria for Evaluation of Instrument	Rating	Justification
Concise		
Sequential		
Valid and Reliable		
Easily Tabulated		

Activity 2. Search and evaluate the instrument.

Directions: Search for a sample research study. Identify the ways of establishing the validity and reliability of the instrument.

	Example 1	Example 2
▪ Title of the Research Study		
▪ Type of Research Instrument		
▪ Way of Developing the Instrument		
▪ Scale Used		
▪ How validity of instrument was established?		
▪ How validity of instrument was established?		
▪ Source		

Activity 3. Let's go online.

Practice what you have learned from this lesson. Visit the following:

<https://bit.ly/2A7Q6lz>

<https://bit.ly/2ZD2Kda>

<https://bit.ly/2B0cbcd>

What I Have Learned



Directions:

As you have learned from this lesson, answer each question comprehensively.

1. Why is it important to have a good research instrument?

2. Differentiate validity and reliability. Explain how they complement each other to make a good research instrument.

What I Can Do

Construct your research instruments and establish validity and reliability



Directions: Fill out the guide table below to be able to create a good research instrument for your study.

1. Goal of your research instrument/s	
2. Parts of your instrument/s	
3. Number of items per subfactor: total items	
4. Scale to be used	
5. How it will be validated?	
6. How will the reliability be established?	



Direction: After you have filled out the guide table above, start constructing your research instrument for your current study.

Title of your study:
Research Questions:
Research Instrument:

4

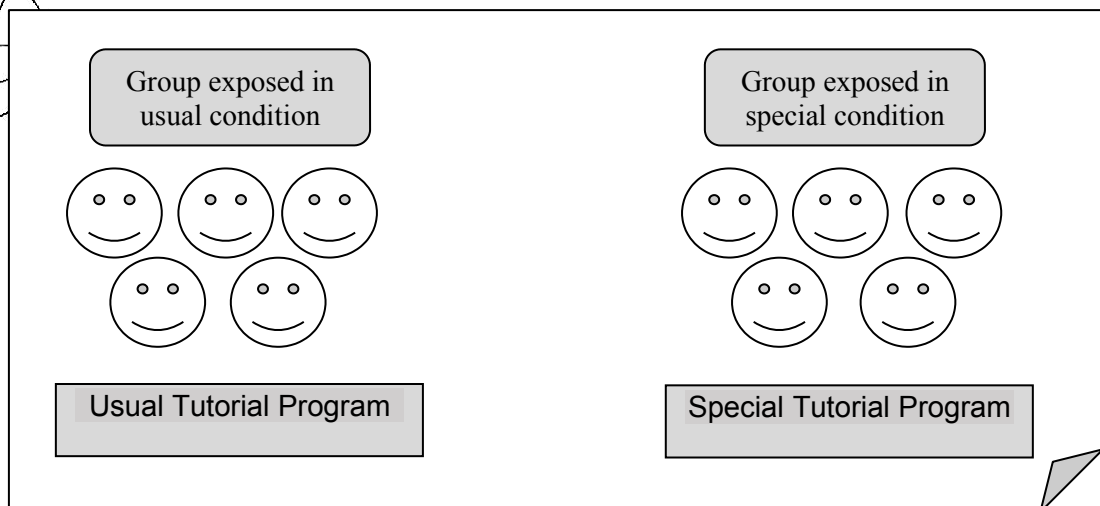
Research Intervention



What's In

In the previous discussion on quantitative research designs, you were taught about quasi-experimental and experimental designs. Its uniqueness from other research designs was also described. Remember that experimental research design controls or manipulates the independent variable. This is done by applying particular conditions or treatment or what is called research intervention. In this lesson, the focus is on how to describe your research intervention in your research paper.

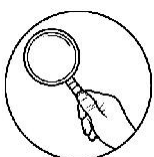
What's New



How is the research intervention conducted in this study?



<https://bit.ly/2YleQqq>



What Is It

Nature of Research Intervention

In experimental research, researcher decides the nature of intervention or treatment. Intervention pertains to what is going to happen to the subjects of the study. This decision covers **who will receive the intervention** and **to what extent** it will be applied to them.

For example in a study of determining the effects of special tutorial program to learners' at risk of failing (LARF), researcher decides the group of LARF who will receive intervention. In this example, special tutorial program is the research intervention. Furthermore, the extent of which the program will be administered to the learners is determined.

Steps in Describing the Research Intervention Process

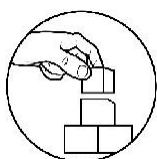
A section that explains the details of research intervention is added in the research paper if it is an experimental design. In this section, procedure of research intervention is explained clearly.

Write the Background Information. It is an introductory paragraph that explains the relevance of the intervention to the study conducted. It also includes the context and duration of the treatment.

Describe the Differences and Similarities between the Experimental and Control Group. State what will happen and what will not both in experimental and control group. This will clearly illustrates the parameters between the research groups.

Describe the Procedures of the Intervention. In particular, describe how will the experimental group receive or experience the condition. It includes how will the intervention happens in order to achieve the desired result of the study. For example, how will the special tutorial program will take place.

Explain the Basis of Procedures. The reason for choosing the intervention and process should clear and concrete reasons. The researcher explains why the procedures are necessary. In addition, theoretical and conceptual basis for choosing the procedures is presented to establish the validity of the procedures.



What's More

Activity 1. Determine the Research Intervention Procedure

Directions: Search for a sample research study. Distinguished the procedure used in experimental and control group. Use the table below for your answers.

	Example 1
▪ Title of the Research Study	

▪ Research Intervention Conducted	
▪ Procedure Used in Control Group	
▪ Procedure Used in Experimental Group	

Activity 3. Let's go online.

Visit the following link and learn further about experimental research.

<https://bit.ly/2Xr5zes>

What I Have Learned



Directions: As you have learned from this lesson, answer each question comprehensively.

1. What I know about research intervention

2. I have learned that when describing the research intervention....

What I Can Do



Describes your Intervention (if applicable)

Directions: If the design of your research is experimental. Describe your research intervention by following the steps given in this lesson.

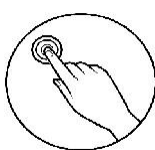
5

Planning Data Collection Procedure



What's In

You have learned the important factors in developing your research instruments in the lesson 3 of this Module. You can now identify the steps you are going to undertake in your actual gathering of data. In this lesson, three phases in data collection will be presented so that you can plan clearly your data collection procedure in your current research.



What I Need to Know

Quantitative Data

Generally, data are any pieces of information or facts that people have known. Once these data answers research problem it becomes helpful to research. When research data appears to be measurable in numerical form it is considered a **quantitative data**. However, some qualitative data can also be useful to quantitative research once it is given numerical value. For example, if you study about adjustment experiences of students to distant learning, if it is categorized and numbered accordingly then it can be quantified during analysis.

Techniques in Collecting Quantitative Data

The following are the common quantitative data gathering technique. Each technique corresponds to specific instrument which will be further discussed in Module 5.

Observation. It is gathering information about a certain condition by means of senses. The researcher records the observation as seen and heard. This is done by direct observation or indirect observation by the use of gadgets or apparatus. An observation checklist aid the researcher in recording the data gathered.

Survey. Data gathering is done through interview or questionnaire. By means of questionnaire you use series of questions or statements that respondents will have to answer. Basically, respondents write or choose their answer from given choices. On the other hand, interview is when you ask respondents orally to tell you the responses. Since you are doing quantitative research, it is expected that responses have numerical value either it is nominal or ordinal in form.

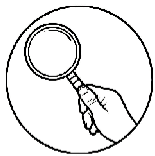
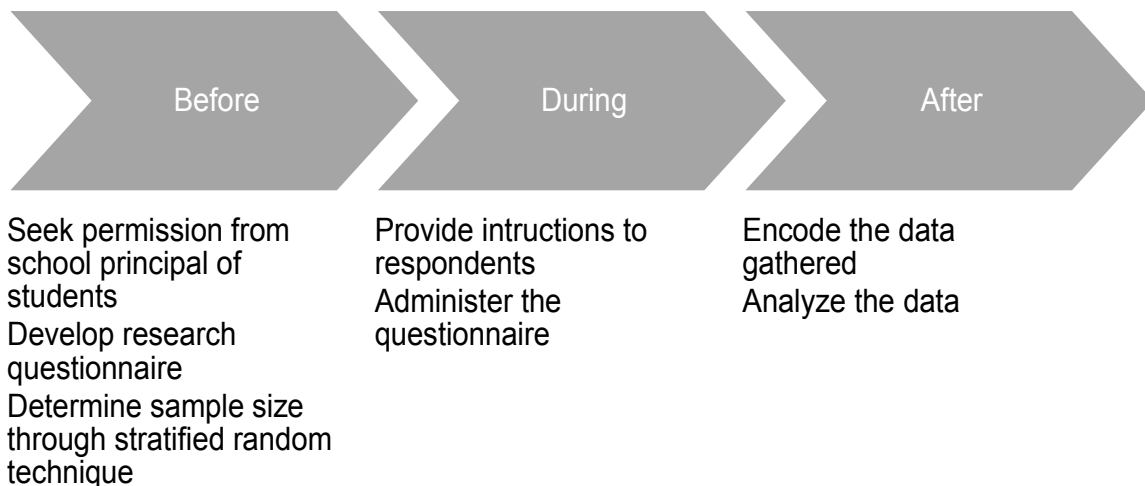
Experiment. When your study is experimental design, it was already discussed in the previous lesson that it will use treatment or intervention. After the chosen subjects, participants, or respondents undergone the intervention the effects of such treatment will be measured.



What's New

Activity 1: What's the Procedure

Looking at the flowchart of data gathering procedure, what do think is the research design of the study?

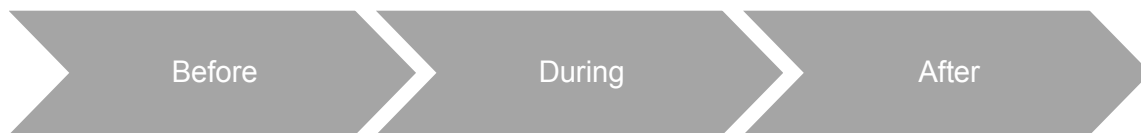


What Is It

Three Phases in Data Collection

In doing research, data collection is a major component of research. Neglecting to clarify the collection procedure would result to acquiring inaccurate data that will make you research study invalid. Hence, data collection procedure is given meticulous attention in order to gather appropriate data. You are making sure that data you will gather answers your research questions.

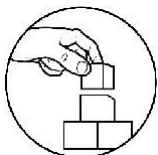
The data gathering procedure is presented in a paragraph format in your research paper. Basically the contents are the steps you are going to follow are: (1) **before** you will gather the data, (2) what to do **during** the actual gathering of data, and (3) the things to consider **after** data has been gathered. The following are the suggested steps but not limited to it, are the procedures in gathering quantitative data.



Prepare the research instruments
 Identify the authorities that will be involved and need to ask permission
 Determine the samples size and corresponding respondents; per group if applicable.
 Ask consent form (if respondents are 18 years old above) or parent's consent (if minor).
 Pilot test the research instrument if needed.

Clear the instructions provided to the respondents.
 Administer the research instrument or implement the research intervention, if applicable.
 Collect or gather or take note of the responses.

Summarize the data gathered, in a tabular form
 Analyze the summarize data corresponding to the research questions.

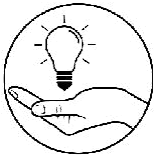


What's More

Activity 2. Arrange your Steps

Directions: Arrange the following steps in data gathering into their correct sequence, 1 as being the first step and 10 as the last step.

Order	Steps
_____	The respondents will have a chance to have a look of the performance tasks.
_____	Each of the participants will be interviewed for individual filling out of personal background of media literacy information.
_____	The researcher will meet the participants eight times in four-month data gathering period.
_____	The participants will answer the learning activity sheet (LAS) after practicing the new approach.
_____	The data collection in each group will be supervised by two experienced Information Technology teacher to ensure proper implementation of the intervention.
_____	The participants will read example situations about media literacy that are not familiar to them.
_____	The new approach in assessing performance tasks of learners will be implemented.
_____	The list of performance tasks and assessment tools will be prepared.
_____	The media literacy summative test will be administered after the implementation of the new approach in assessing performance tasks.
_____	The test results will be encoded for summary and will be analyzed.



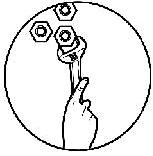
What I Have Learned

Directions: As you have learned from this lesson, answer each question comprehensively.

1. This time, I have learned that quantitative data.....

2. Techniques in quantitative data gathering helps to.....

3. In planning my data gathering procedure I need to consider.....



What I Can Do

Plan your Data Gathering Procedures

Directions: Perform the following task. From what you have learned in this lesson, list the steps that you are going to follow in gathering the data of your research. For this part, just enumerate it to clearly see the procedure before, during, and after your data gathering.

Before:

During:

After:



Additional Activity:

Direction: After you submit your data gathering procedures, once it is corrected in accordance to the guidelines given, you may start incorporating it to your research manuscript in paragraph format.

Lesson 6

Planning Data Analysis



What's In

Prerequisite for taking Practical Research 2 is your Statistics and Probability subject. It is presumed that you have already a good practice of the learning competencies needed to conduct quantitative research. Your background statistics and probability background will help you plan and choose your data analysis.

In planning your data analysis in quantitative research, you need also to consider your research problem, type of data, hypothesis, and scale used in your research instrument. This lesson focuses on designing your data analysis procedure.



What I Need to Know

Data Analysis

Data analysis in research is a process in which gathered information are summarized in such a manner that it will yield answers to the research questions. During quantitative data analysis gathered information were break down and ordered into categories in order to draw trends or patterns in a certain condition. In quantitative research, the numerical data collected is not taken as a whole. In order to understand it better, it is analyze into components based on the chosen research variables and research questions you are going to answer.

These numerical data are usually subject to statistical treatment depending on the nature of data and the type of research problem presented. **Statistical treatment** makes explicit the different statistical methods and formulas needed in order to analyze the research data.

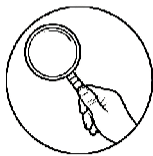


What's New

Activity 1: Statistics Recall

Directions: From your previous lessons, recall the following terms in Statistics. Write its definition or description, and purpose or function in analyzing data. Fill in the boxes in order to provide the information about these statistical terms. Use the table for your answer.

Terms	Definition or Description	Purpose or Function
Data		
Frequency		
Percentage		
Mean		
Standard Deviation		
Table		
Figure		
Parametric Test		
Non-Parametric Test		
Correlation		
Regression		



What Is It

Planning your Data Analysis

Before choosing what statistical test is appropriate for your research study it is important to determine what statistical formation is applicable to your current study. In immersing yourself into planning your data analysis, you have to decide what basic descriptive statistical technique you are going to use. Although this technique does not give you the degree of association or effect between variables, this will help you to code and simply tabulate your data.

Descriptive Statistical Technique provides summary of the ordered or sequenced data from your research sample. Frequency distribution, measure of central tendencies (mean, median, mode), and standard deviation are the sets of data from descriptive statistics. **Inferential Statistics** is used when the research study focuses on finding predictions; testing hypothesis; and finding interpretations, generalizations, and conclusions. Since this statistical method is more complex and has more advanced mathematical computations, you can use computer software to aid your analysis.

You also have to identify types of statistical analysis of variable in your quantitative research. A **univariate analysis** means analysis of one variable. Analysis of two variables such as independent and dependent variables refers to **bivariate analysis** while the **multivariate analysis** involves analysis of the multiple relations between multiple variables.

Furthermore, selecting what test to use is basically done by identifying whether you will use parametric test or non-parametric test. As these were already discussed in your Statistics and Probability subject, a summary of what to consider is presented below:

Points to Consider		Type of Test
Scale	Interval or Ratio	Parametric Tests
	Ordinal or Nominal Scale	Non-parametric Tests
Sample Size	30 or more per group	Parametric Tests
	Fewer than 30	Non-parametric Tests
Distribution of Data	Normal Distribution	Parametric Tests
	Data deviates markedly	Non-parametric Tests

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In addition, in choosing statistical techniques in quantitative research the purpose or objective of the research study should be considered.

Test of Relationship between Two Variables

- Pearson's r (parametric)

- Phi coefficient (non-parametric for nominal and dichotomous variables)
- Spearman's rho (non-parametric for ordinal variable)

Test of Difference between Two Data Sets from One Group

- T-test for dependent samples (parametric)
- McNemar change test (non-parametric for nominal and dichotomous variables)
- Wilcoxon signed-rank test (non-parametric for ordinal variable)

Test of Difference between Two Data Sets from Two Different Groups

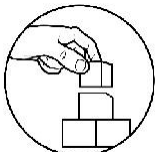
- T-test for independent samples (parametric)
- Two-way chi-square (non-parametric for nominal variable)
- Mann-Whitney U test (non-parametric for ordinal variable)

Test More than Two Population Means

- Analysis of Variance or ANOVA (parametric)

Test the Strength of Relation or Effect or Impact

- Regression (parametric)



What's More

Activity 2. Step by step analysis

Directions: From the concept presented in this lesson, create five simple basic procedures in planning the data analysis. What to do beforehand (pre-process) is already given as your guide.

Step	To Do
Pre-processes	Identify the types of variables, research question, hypothesis, and scale of measurement.
1.	
2.	
3.	
4.	
5.	

Activity 3. Choose the Appropriate Statistical Test

Directions: Determine the statistical test/s appropriate for the sample research. Make sure to explain your decision in two to three sentences.

1. Relationship between Academic Stressors and Learning Preferences of Senior High School Students

Statistical Test/s: _____

Explanation:

2. Reading Electronic Learning Materials as a Support for Vocabulary of Grade 1 Pupils

Statistical Test/s: _____

Explanation:

3. Impact of the Implementation of COVID – 19 Health Protocols in Supermarkets on Consumer Behaviors

Statistical Test/s: _____

Explanation:

4. Effects of Morning Exercise on the Health Anxiety Level of Senior Citizens

Statistical Test/s: _____

Explanation:

5. Measuring the Gadgets Usage of Children at Home during Quarantine

Statistical Test/s: _____

Explanation:

Activity 4: Let's go Online

Go to the link below to learn more about statistical techniques for quantitative research.

<https://bit.ly/3c6evv2>
<https://bit.ly/36EfqbZ>
<https://bit.ly/3gmMbYS>
<https://bit.ly/2ZGYNEu>
<https://bit.ly/2A7L6DE>
<https://bit.ly/2TGK4Fr>

What I Have Learned



Directions. As you have learned from this lesson, answer each question comprehensively.

1. This time, I have learned that quantitative research and statistics.....

2. The factors to consider before planning the data analysis of research study are.....

3. How do you know the specific statistical tests to be used in a research study?

What I Can Do

Plan Data Analysis Using Statistics and Hypothesis Testing



Directions: Perform the following task. From what you have learned in this lesson, fill out the table considering your current research study.

Research Questions	Statistical Tests	Purpose



Additional Activity:

Direction: After you submit your data analysis plan, once it is corrected in accordance to the guidelines given, you may start incorporating it to your research manuscript in a paragraph format.

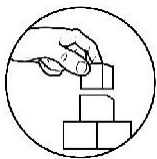


What Is It

Indeed, designing the research methodology in quantitative research is quite challenging. At this point, it is assumed that you are now ready to present your written output. You need to consider the parts of your research methodology, these are:

- Research Design
- Research Population and Sample
- Sampling Procedure
- Research Instruments
- Validity and Reliability of Instruments
- Research Intervention (if applicable)
- Data Collection Procedure
- Data Analysis

All of these are written in paragraph format as part of your research methodology. In this lesson, you will be given guidelines in presenting this research portion. After presentation, the most exciting part follows; and that is gathering your data.



What's More

Activity 1. Personal Work Evaluation

Direction: Rate your own paper using the rubric below. For the purpose of improvement, rate your output as honestly as you can. Use the following scale in evaluating your own paper.

5 – Very Good, 4 – Good, 3 – Fair, 2 – Poor, 1 – Needs Improvement

Indicator	Self-Rating
1. Research Design <i>Type and definition were presented, and rational is discussed.</i>	
2. Sampling Procedure and Sample <i>Population and samples are described, and sampling procedure is clear and relevant.</i>	
3. Research Instrument, and its Validity and Reliability <i>Basis of decision is explained, instrument is well described, and its validity and reliability were clearly presented.</i>	
4. Data Gathering Procedure <i>Protocols were clear, and steps in gathering are well stated.</i>	
5. Data Analysis <i>Basis and explanation are clearly discussed.</i>	
Total Rating	

Activity 2. Reflection

1. What is your highest and lowest rating? What are your reasons for giving yourself such ratings?

2. Overall, do you think you are ready for presentation of your research paper? Explain your answer.

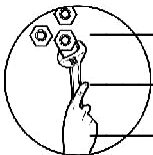
Activity 3. Let's go online

Visit the following links to further enhance your written problem and to improve your preparation for presentation.

<https://bit.ly/2TFhfcy>

What I Have Learned

1. What are the new learning you have gained from this module in terms of skills, content, and attitude in identifying and stating the research problem?



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2. How will you improve your research writing skills in quantitative research?

What I Can Do

Direction: Get ready for presentation of your written statement of the problem. The rubrics below will served as a guide on how you will be rated by your teacher. The 4C (content, coherence, creativity, communication) technique will be used so that you can easily remember.



Rubrics for Written Statement of the Problem

	5	4	3	2	1
Content (35%) ▪ The parts of the paper are complete and appropriate. ▪ The parts of the research methodology are clearly described and justified. ▪ Overall content is comprehensive.					
Coherence (25%) ▪ Cohesive of devices are effectively used. ▪ Organization of ideas is smoothly presented.					
Creativity (25%) ▪ Writer's voice is showcased. ▪ Paper uses variety of sentence structures. ▪ Uses appropriate language.					
Communication (15%) ▪ Sentences are well structured ▪ Grammatical conventions are observed. ▪ Correct spelling and proper research format is followed.					

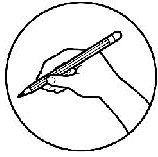
Legend: 5 – to a very great extent, 4 – to a great extent,
3 – to some extent, 2 – to a little extent, 1 – not at all

Summary

- In order to integrate all the components of the research study, the **research design** describes the overall strategy of the research. **Types of quantitative research** are descriptive, correlational, ex post facto, quasi-experimental, and experimental research.
- **Population** refers to all the members of a particular group relevant to the research. **Sample** is a part of the population that serves as a representative. Sample are chosen through **sampling** which is a process of selecting of who will participate. The usual

sampling procedures in quantitative research are simple random sampling, stratified random sampling, cluster sampling, and systematic sampling.

- **Research Instrument** refers to the tools used in research for the purpose of gathering the data. **Common scales** used in quantitative research are Likert Scale and Semantic Differential.
- It is important in quantitative research to use valid instruments. **Validity** refers to the appropriateness of the research instrument. **Types of validity** are face, content, construct, concurrent, and predictive validity.
- The consistency in research findings provided by instruments refers to **reliability**. **Types of reliability** are test-retest, equivalent forms, and internal consistency.
- For quasi-experiment and experimental research designs, **intervention** is applied in order to know who will receive the intervention and to what extent.
- **Data gathering** in quantitative research usually observes survey, observation, and experiment. There are **three phases in data gathering** of research that need to be presented in the research paper: before, during, and after procedures.
- **Data Analysis** is a process in which gathered information are summarized in such a manner that it will yield answers to the research questions. There are two major statistical technique in data analysis, these are descriptive and inferential statistics.



Assessment: (Post-Test)

Directions: Read and analyze the statements below. Encircle the letter of the correct answer.

1. Ex post facto research design aims to:
 - A. determine a cause from already existing effects.
 - B. identify association among variables
 - C. observe and describe a phenomenon
 - D. establish cause and effect relationship
2. "Effects of Type of Music to Aesthetic Performance of Ballet Dancers", the appropriate research design for the given title is
 - A. Correlational
 - B. Descriptive
 - C. Survey Research
 - D. Experimental
3. Mr. Canuc would like to further know the type of social media used between the male and female SHS students of East Pagat National High School. What is the appropriate research design to be used in his study?
 - A. Correlational
 - B. Quasi-Experimental
 - C. Descriptive
 - D. Experimental
4. Based on item number 3, what appropriate statistical test should Mr. Canuc used to answer his research problem?
 - A. T-test for two dependent samples
 - B. Spearman's rho
 - C. Chi-square
 - D. ANOVA
5. Which of the following statements is true about the conduct of experimental research?
 - A. Intact groups are used.
 - B. There is no random assignment of individuals.
 - C. Individual subjects are randomly assigned.
 - D. Groups are exposed to presumed cause.
6. What is the difference between quasi-experimental research and experimental research?
 - A. Intact groups are used in experimental, while quasi-experimental randomly assigned individuals into groups.
 - B. The researcher controls the intervention in the experimental group in experimental, but not quasi-experimental research.
 - C. Only one dependent variable is used in quasi-experimental research, while multiple dependent variables can be used in quasi-experimental research.
 - D. Participants for control and experimental groups are randomly selected in experimental, but not quasi-experimental research.
7. Simple Random Sampling is used when the researcher would like to
 - A. consider giving equal chance to the member of accessible population being selected as part of the study.
 - B. make sure that all subcategories of the population are represented in the selection of sample.
 - C. group the entire population into clusters since the location of the samples are widely spread.
 - D. systematically choose samples from a given list of individuals.

8. The best research sample is when it is
- A. purposely selected
 - B. systematically chosen
 - C. representative of population
 - D. conveniently represented
9. Given that your study will use stratified random sampling, wherein population of your scope is 250 with a computed sample size of 152, how many samples for each strata will you have if group 1 has 92, group 2 has 86, and group 3 has 72 population?
- A. Group 1 = 52, Group 2 = 54, Group 3 = 46
 - B. Group 1 = 56, Group 2 = 45, Group 3 = 51
 - C. Group 1 = 52, Group 2 = 52, Group 3 = 44
 - D. Group 1 = 54, Group 2 = 56, Group 3 = 41
10. It is a type of reliability when measured by administering two tests identical in all aspects except the actual wording of items.
- A. Equivalent Forms Reliability
 - B. Internal Consistency Reliability
 - C. Test-retest reliability
 - D. Inter-rater Reliability
11. It is a type of validity when an instrument produces results similar to those of another instrument that will be employed in the future.
- A. Face Validity
 - B. Predictive Validity
 - C. Criterion Validity
 - D. Content Validity
12. The Ability Test has been proven to predict the mathematical skills of Senior High School students is an example of
- A. Face Validity
 - B. Construct Validity
 - C. Content Validity
 - D. Criterion Validity
13. It is a good description of research instrument when items are arranged from simple to complex.
- A. Concise
 - B. Sequential
 - C. Valid and Reliable
 - D. Easily Tabulated
14. The purpose of Pearson's r as a statistical technique is to test the
- A. relationship between two variables.
 - B. degree of effect research intervention or treatment.
 - C. difference between two sets of data from one group.
 - D. difference between sets of data from different groups.
15. What statistical technique should be used for this research problem, "You would like to determine the differences between the opinions of men and women on the COVID – 19 local government response."
- A. T-Test for two dependent samples
 - B. T-Test for independent samples
 - C. Two-way Chi-square
 - D. Pearson's r



Key to Answers

Pretest

1. C
2. D
3. B
4. B
5. D
6. A
7. C
8. B
9. C
10. A
11. C
12. B
13. C
14. B
15. C

Posttest

1. A
2. D
3. A
4. C
5. A
6. D
7. A
8. C
9. C
10. A
11. B
12. D
13. B
14. A
15. B

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